

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Compute XAI Agreement Metrics
2. Abbreviated Title	XAI Agreement
3. Use Case Id	UC-18
4. Actors	ML Engine (Supporting)
5. Description	The system computes agreement and consistency metrics across the three XAI methods (Grad-CAM, SHAP, Attention) to quantify how much they agree on which clinical features are most important, boosting explanation reliability.
5.1. Pre Conditions	1. All three XAI outputs (UC-12, UC-13, UC-14) are available. 2. Feature importance rankings from each method are normalized.
5.2. Task Sequence	1. System normalizes the feature importance rankings from each XAI method. 2. System computes pairwise rank correlation (Spearman's rho). 3. System computes top-k feature overlap across all 3 methods. 4. System computes an overall agreement score. 5. System outputs the score and a visual agreement matrix.
5.3. Post Conditions	1. An agreement score is computed and displayed on the dashboard. 2. Clinician can assess the reliability of XAI explanations.
6. Modification History	Date: September 11, 2026
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